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25 FEBRUARY 1950
SUPERSEDING
ARMY—50-21-38A
31 DECEMBER 1947

MILITARY SPECIFICATION

SHELL, FIXED, CHEMICAL, M64 FOR 75-MM. GUNS, M312 FOR 76-MM. GUNS, M313 FOR 90-MM. GUNS, AND SHELL, SEMIFIXED, CHEMICAL, M64 FOR 75-MM. HOWITZERS, M60 FOR 105-MM. HOWITZERS, ASSEMBLING, MARKING, AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments

1. SCOPE

1.1 Scope.—This specification covers assembling, marking, and packing of fixed chemical shell for 75-, 76-, and 90-mm. guns and semifixed chemical shell for 75- and 105-mm. howitzers.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.—Specifications referenced on the applicable drawings (see 2.2), of the issue in effect on date of invitation for bids, form a part of this specification.

2.2 Drawings.—The following drawings, and all drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

ORDNANCE DEPARTMENT DRAWINGS

75-1-108—Shell, Semifixed, Smoke, FS, M60, Dualgran, w/fuze, PD, M51A5, 0.05 Sec. Delay, for 105-mm. Howitzers, M2A1 and M4.

75-1-109—Shell, Semifixed, Gas, Persistent, H, M60, Dualgran, w/fuze, PD, M51A5, 0.05 Sec. Delay, for 105-mm. Howitzers, M2A1 and M4.

75-1-110—Shell, Semifixed, Smoke, WP, M60, Dualgran, w/fuze, PD, M51A5, 0.05 Sec. Delay, for 105-mm. Howitzers, M2A1 and M4.

75-1-111—Shell, Fixed, Smoke, WP, M64, w/fuze, PD, M48A3, 0.05 Sec. Delay, for 75-mm. Guns.

75-1-114—Shell, Semifixed, Smoke, WP, M64, w/fuze, PD, M48A3, 0.05 Sec. Delay, for 75-mm. Howitzers.

75-1-116—Shell, Semifixed, Smoke, FS, M64, w/fuze, PD, M48A3, 0.05 Sec. Delay, for 75-mm. Howitzers.

75-1-224—Shell, Fixed, Smoke, WP, M313, Smokeless, w/fuze, PD, M48A3, 0.05 Sec. Delay, for 90-mm. Guns, M1, M1A1, M2, M2A1, M3, M3A1, and T8.

75-1-227—Shell, Fixed, Smoke, WP, M312, Flashless-Smokeless, w/fuze, PD, M48A3, 0.05 Sec. Delay, for 76-mm. Guns.

(Copies of specifications and drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Materials.—Materials and parts shall meet all requirements specified on applicable drawings and in applicable specifications.

3.2 Rotating band.—The rotating band of the projectile shall not be damaged, for example, nicked or burred.

3.3 Painting and marking.—The painting and markings shall be restored by the assembling contractor on any components furnished to him painted and marked, when such painting and markings have been marred or obliterated.

3.4 Propelling charge (rounds for guns only).

3.4.1 *Rounds for 75-mm. guns.*—The weight of the propelling charge shall not differ by more

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than ±0.03 ounce avoirdupois from that designated by the loading authorization for the particular lot of powder used. (See 3.4.4.)

3.4.2 *Rounds for 76-mm. guns.*—The weight of the propelling charge shall not differ by more than ±0.05 ounce avoirdupois from that designated by the loading authorization for the particular lot of powder used. (See 3.4.4.)

3.4.3 *Rounds for 90-mm. guns.*—The weight of the propelling charge shall not differ by more than ±0.09 ounce avoirdupois from that designated by the loading authorization for the particular lot of powder used. (See 3.4.4.)

3.4.4 *Weighing of charge.*—The contractor shall weigh and checkweigh each propelling charge. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that used to make the first weighing.

3.5 **Crimp.**

3.5.1 *Rounds for 75- and 76-mm. guns.*—Cartridge cases shall not separate from the shell at a load of less than 1,800 pounds.

3.5.2 *Rounds for 90-mm. guns.*—Cartridge cases shall not separate from the shell at a load of less than 5,000 pounds.

3.6 **Proving ground test.**

3.6.1 *Average velocity.*—The difference between the velocities of any two rounds fired in the velocity test shall not exceed 3 percent of the mean velocity of all the rounds fired in this phase of the test. The average velocity shall not differ from the prescribed velocity by more than ±2 percent after reducing results to standard conditions.

3.6.2 *Premature bursts.* — No premature bursts or evidence thereof shall occur.

3.6.3 *Shell functioning.*—In any ballistic sample fired, not more than one shell shall fail to function satisfactorily.

3.7 **Workmanship.**—All parts shall be free from chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Round lots.**

4.1.1 *Lot size.*—Maximum lot size shall be 25,000. Minimum lot size shall be 2,500.

4.1.2 *Component limitations.*—Each ammunition lot shall be limited to:

4.1.2.1 Cartridge cases of new manufacture limited to one manufacturer, same drawing and drawing revision, and same specification and specification revision.

4.1.2.2 Filled shell of the same weight zone which shall contain:

4.1.2.2.1 Shell made by one manufacturer at one plant, in accordance with the same drawing and drawing revision, and same specification and specification revision.

4.1.2.2.2 Burster charges from not more than two lots or two grand lots (if grand lot assignments have been made for the item). These lots should be consecutive lots if possible (not a requirement).

4.1.2.2.3 Burster initiators from not more than two lots or two grand lots (if grand lot assignments have been made for the item). These lots should be consecutive lots if possible (not a requirement).

4.1.2.3 Loaded fuzes from not more than two lots or two grand lots (if grand lot assignments have been made for the item). These lots should be consecutive lots if possible (not a requirement).

4.1.2.4 Loaded primers from not more than two lots or two grand lots (if grand lot assignments have been made for the item). These lots should be consecutive lots if possible (not a requirement).

4.1.2.5 Propellent powder of the same lot.

4.2 **Sampling.**—Samples shall be uniformly selected from each round lot as follows:

4.2.1 Select 10 samples for test of crimping case to shell. (See 4.4.)

4.2.2 Select 20 sample rounds for proving ground tests. (See 4.5.)

4.3 **Inspection.**

4.3.1 *Rounds for guns and howitzers.*—Inspection shall be as specified in Specification 50-0-1.

4.3.2 *Gaging (rounds for guns).*—After the round has been assembled, the contractor shall gage each round of ammunition in a profile and alinement gage (commonly known as a chamber gage) pertaining to this round. If the round fails to enter the profile and alinement gage freely, it shall be rejected subject to correction and retest. Drawings of the Government pro-

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file and alinement gage will be furnished to the contractor for his information upon application to the inspector.

4.4 **Test of crimping case to shell (rounds for 75-, 76-, and 90-mm. guns).**—Sample rounds selected as prescribed in 4.2.1 shall be tested as follows:

4.4.1 *Procedure.*—Use a tensile testing machine satisfactory to the inspector. Determine the security of the crimp of the cartridge case to the shell by applying a load in the axial direction. Rate of travel of testing machine head shall be 0.100 to 0.125 inch per minute inclusive.

4.4.2 *Acceptance.*—The lot will be considered acceptable provided that none of the 10 samples fails to meet the requirements of 3.5. If only one sample fails to meet this requirement, the lot will be considered eligible for retest under the provisions of 4.4.3. If more than one sample fails to meet this requirement, the lot shall be rejected.

4.4.3 *Retest.*—A retest as provided for in 4.4.2 may be made, using 20 sample rounds for the retest. The lot will be considered acceptable provided that none of the samples in the retest fails to meet the requirements of 3.5.

4.5 **Proving ground tests.**—Sample rounds selected as prescribed in 4.2.2 shall be tested as follows:

4.5.1 *Velocity.*—Fire five of the sample rounds with fuzes set "SQ" action, in a weapon for which the round is standard (with full charge in a howitzer), for impact on firm ground, and measure the velocities. Fuzes shall not be fired through velocity screens except solenoid coils. Observe for premature bursts and for size and character of the smoke cloud.

4.5.2 *Safety.*—Disassemble five of the sample rounds and reassemble with powder charges adjusted to give pressures 112 percent of the average pressure given by the propelling charge loaded in the particular ammunition lot. This value of pressure appears on the data card of each ammunition lot. Fire in a weapon for which the round is standard, at any convenient weapon elevation, with fuzes set "SQ" action, and make observations for premature bursts and for size and character of the smoke cloud.

4.5.3 *Functioning tests.*—Fire the remaining sample rounds in a weapon for which the round is standard (with full charge in a howitzer), as follows:

4.5.3.1 Five rounds with fuze set "superquick" action, at any convenient weapon elevation, for nose impact on firm ground, and observe for functioning and size and character of smoke cloud.

4.5.3.2 Five rounds with fuze set "delay" action, for impact against 4-inch wood screen placed not less than 400 feet from the muzzle of the weapon, and observe for location of functioning, and for size and character of smoke cloud.

4.5.4 *Acceptance.*—The lot will be considered acceptable provided that no premature burst occurred and 19 of the 20 shell functioned satisfactorily. If no premature burst occurred and 18 of the 20 shell functioned satisfactorily, the lot will be considered eligible for retest under the provisions of 4.5.5. If a premature burst occurs or less than 18 of the 20 shell function satisfactorily, the lot shall be rejected.

4.5.5 *Retest.*—A retest as provided for in 4.5.4 may be made if requested by the contractor, using 40 sample rounds for the retest. The lot will be considered acceptable provided that no premature burst occurred and that 39 of the 40 shell in the retest function satisfactorily.

4.6 **H. shell loading.**—When H. shell only are being loaded, the ballistic samples shall be loaded with FS (smoke) so that the proper observation of burst can be made.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.**—Packaging and packing shall be as specified on the drawing.

5.2 **Marking.**—The marking for shipment shall be as specified on the drawing.

5.3 **Data cards.**—Data cards, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted rounds to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

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5.3.1 *Data card forms.*—Forms to be used for data cards shall be as follows, except that headings need be made only for applicable components:

5.3.1.1 *Rounds for guns.*

AMMUNITION DATA CARD

Kind _____ Ammunition lot No. _____
AIC symbol _____ Quantity in lot _____
Round dwg. No.* _____ Specialty No.* _____
Assembled by _____ Date _____
Expected muzzle velocity _____
Weight propelling charge _____ Pressure (pounds per square inch) _____
Contract No. _____ Order No. _____
Packed _____ in a box
Remarks _____

COMPONENTS

	Model	Manufacturer	Lot No.	Year manufactured	Date filled	Method	Drawing No.*
Cartridge case							
Primer							
Propellant							
Shell							
Filler							
Fuze							
Burster initiator							
Burster charge							
Fiber container							
Metal container							
Wood box							

Certified to by _____ inspector. Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable.

*Include revision and amendments.

5.3.1.2 *Rounds for howitzers.*

AMMUNITION DATA CARD

Kind _____ Ammunition lot No. _____
AIC symbol _____ Quantity in lot _____
Round drawing No.* _____ Specification No.* _____
Assembled by _____ Date _____
Expected muzzle velocity _____
Weight propelling charge _____ Pressure, pounds per square inch _____
Contract No. _____ Order No. _____

Zone	Weight charge by zone	Total weight charge	Muzzle velocity	Pressure
1				
2				
3 (Provision shall be made for 7-zone charges for the 105-mm. howitzer complete rounds, and 4-zone charges for the 75-mm. howitzer complete rounds.)				
4				
5				
6				
7				

Packed _____ in a box
Remarks _____

*Include revision and amendments.

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COMPONENTS

	Model	Manufacturer	Lot No.	Year manufactured	Date filled	Method	Drawing No.*
Cartridge case							
Primer							
Propellant							
Shell							
Filler							
Fuze							
Burster initiator							
Burster charge							
Fiber container							
Metal container							
Wood box							

Certified to by ..... inspector. Print here name of inspector, district, arsenal, or loading plant, and subdistrict, where applicable.

*Include revision and amendments.

6. NOTES

6.1 Round.—A round as referred to herein means the assembly of all the components of ammunition into one unit necessary for firing the round from a weapon; example: cartridge case, loaded percussion primer, propelling charge, loaded shell, loaded fuze, and loaded burster initiator, all properly assembled.

6.2 Tests.—When the contractor for the round is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground test of the round to save expense upon agreement between the United States Government and the contractor to that effect. In case the round specification does not cover all of the proving ground tests specified for the components, the additional tests prescribed in the component specifications will be conducted.

6.3 Premature burst.—A "premature burst" is defined as a functioning of the shell in the gun bore or in flight before impact.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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DOVER, N.J.

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PLATE NO. 15419 (BACK)

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